

NIVA SENSING INSOLE FLEX – LEFT MEDIUM – ENGINEERING PROTOTYPE – NOT FOR FABRICATION

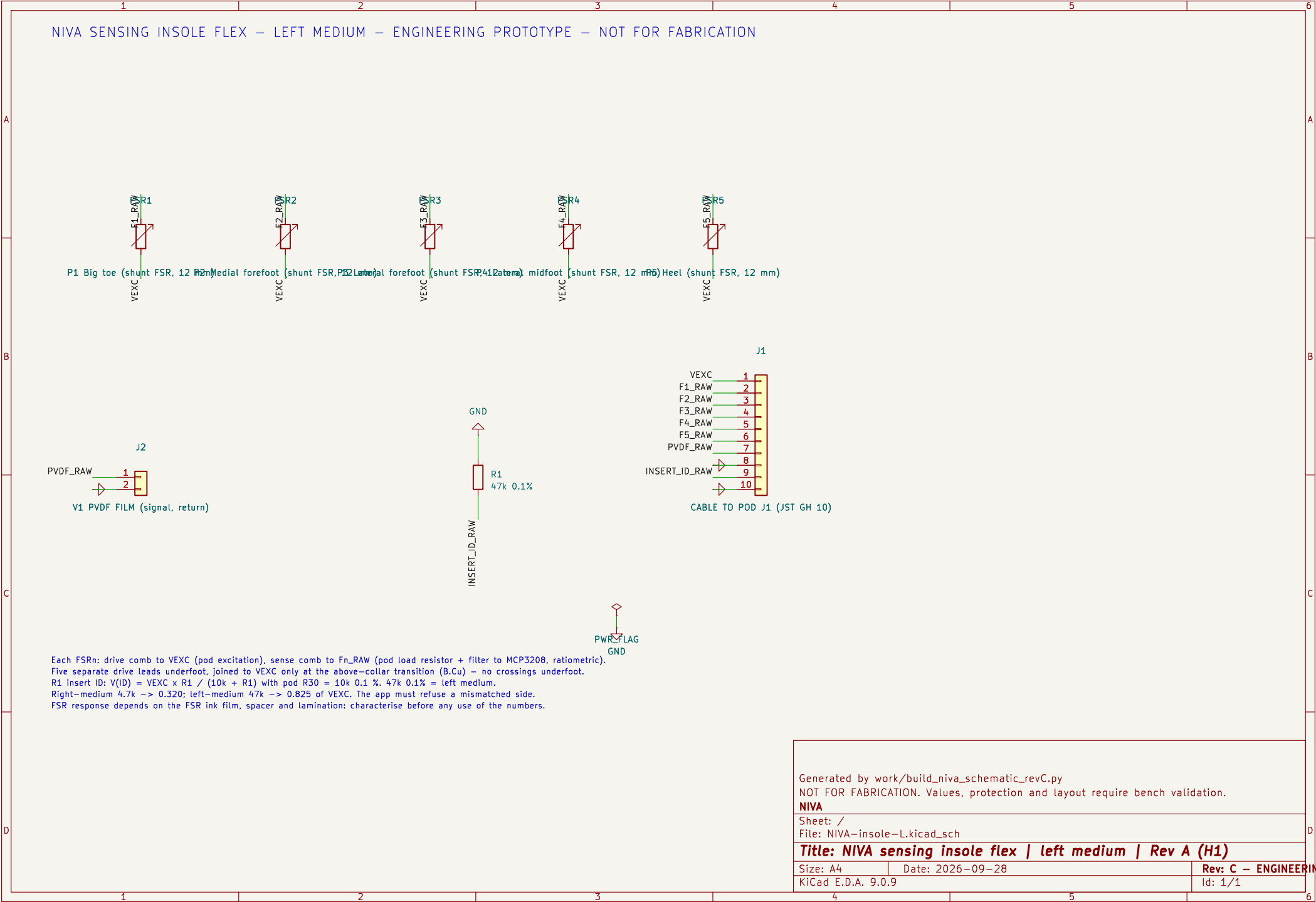
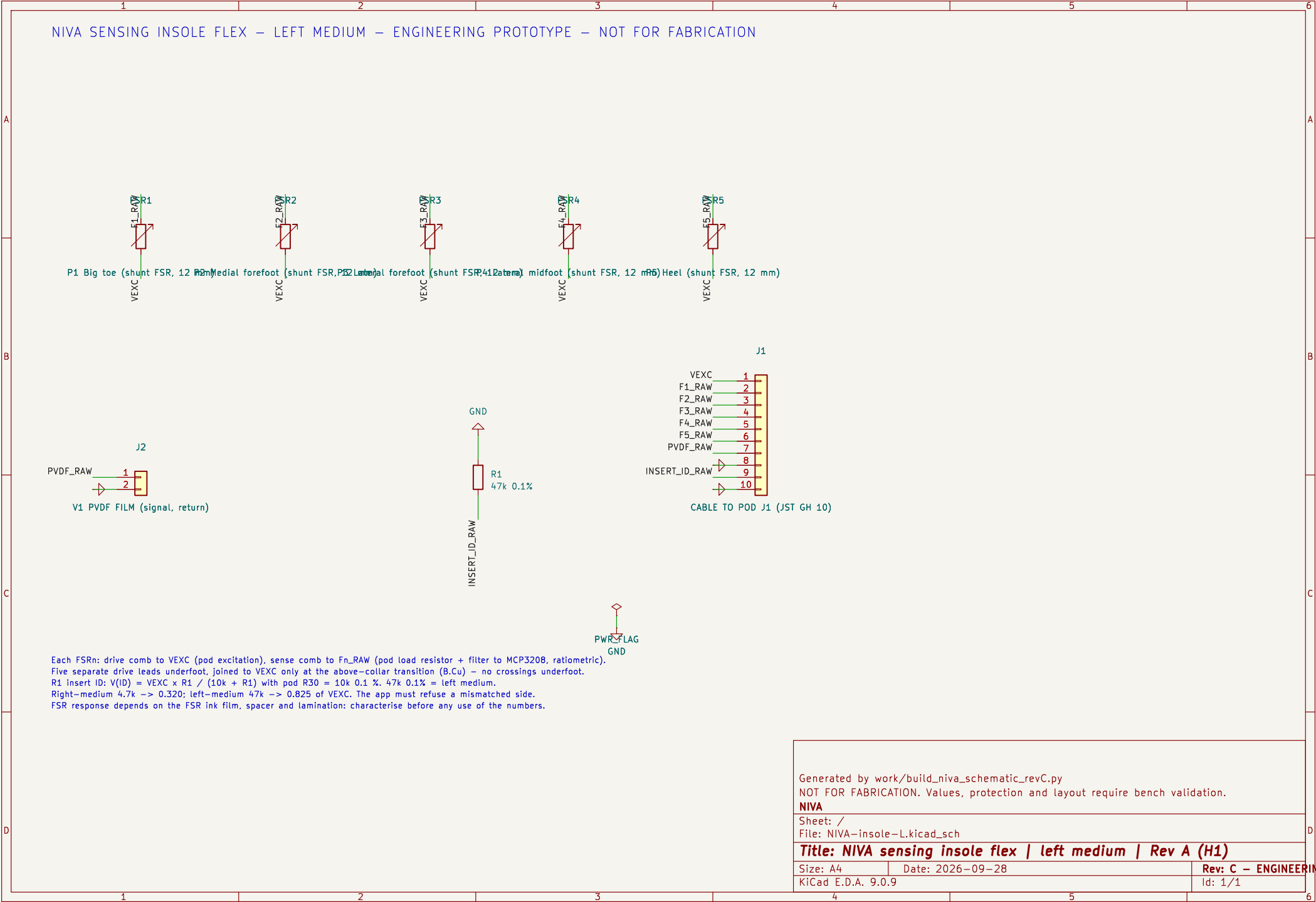
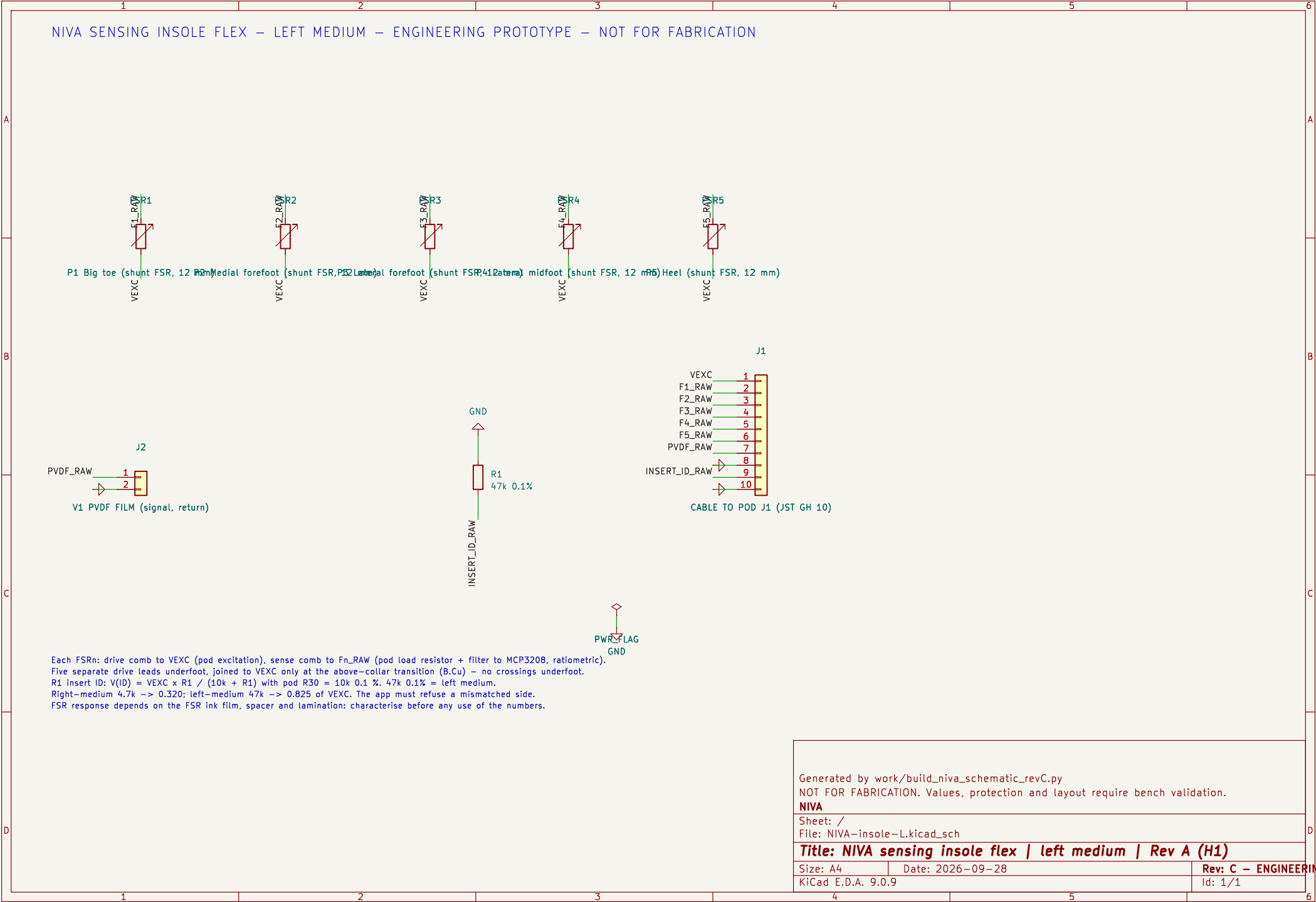
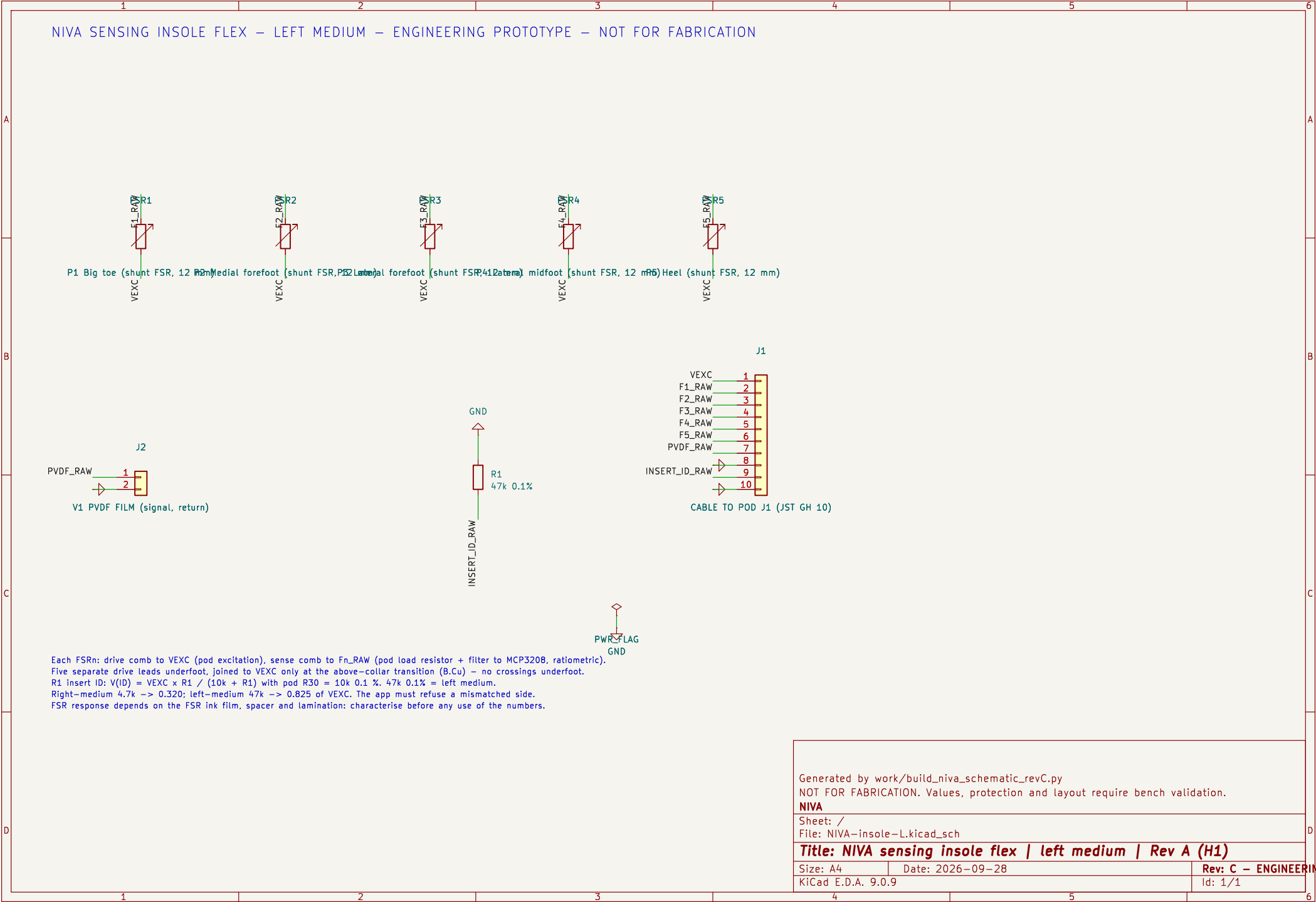
Each FSRn: drive comb to VEXC (pod excitation), sense comb to Fn_RAW (pod load resistor + filter to MCP3208, ratiometric).
Five separate drive leads underfoot, joined to VEXC only at the above-collar transition (B.Cu) – no crossings underfoot.
R1 insert ID: $V(ID) = VEXC \times R1 / (10k + R1)$ with pod $R30 = 10k \ 0.1 \ \%$. $47k \ 0.1\% =$ left medium.
Right-medium $4.7k \rightarrow 0.320$; left-medium $47k \rightarrow 0.825$ of VEXC. The app must refuse a mismatched side.
FSR response depends on the FSR ink film, spacer and lamination: characterise before any use of the numbers.

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.
NIVA
Sheet: /
File: NIVA-insole-L.kicad_sch
Title: NIVA sensing insole flex | left medium | Rev A (H1)
Size: A4 Date: 2026-09-28 Rev: C – ENGINEERING
KiCad E.D.A. 9.0.9 Id: 1/1

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